

EN 601.124

The Ethics of Artificial Intelligence and Automation:
Theory of Change, Institutions, Partnerships, and Policy

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Announcements

- Tomorrow: Reading reflection due
- Next week:
 - **Midterm exam** on Monday Mar 9th
 - **Project proposal presentations** on Wednesday Mar 11th
 - 10 minutes max, overview of the proposed project
 - The goal is to cover the points from the proposal, and receive feedback from the class
 - 2 team members should present

Today's class

- Build our framework such that we move from reasoning about ethics, to: Addressing social problems with AI in an **ethical and responsible** way
- What does it take to do that?
- Objectives:
 - Learn how to specify a problem
 - Identify stakeholders
 - Specify a causal theory of change
 - Identify institutions and partners
 - Perform evaluation and risk assessment

Today's class

- Build our framework such that we move from reasoning about ethics, to:

Addressing social problems with AI in an **ethical and responsible** way

- What does it take to do that?

Framework for Ethical AI Deployment

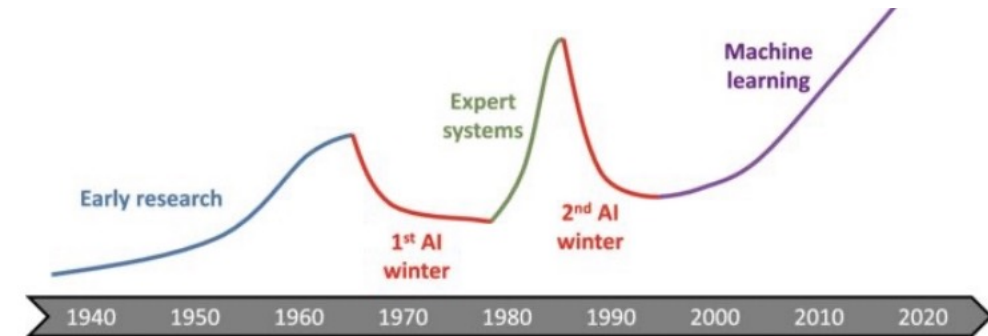
1. Define the social problem
2. Identify stakeholders
3. Specify theory of change
4. Identify institutions & partners
5. Define evaluation metrics
6. Assess risks

1. The role of the ToC

Why this matters?

Many AI ideas fail! Not because of malice or incompetence, but because:

- problem poorly defined
- wrong stakeholders
- unrealistic theory of change
- missing institutional partners
- problem didn't call for an AI solution (bubble and hype dynamics)



AI ethics is not just about AI

Ethical AI requires:

- rigorous problem framing
- institutional awareness
- responsible real-world deployment

Our running example today

Kidney exchange algorithms

Problem:

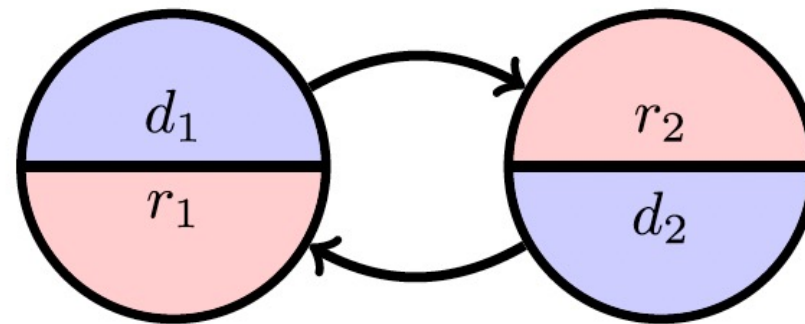
- There are currently more than 96,000 people waiting for a kidney in the U.S., according to data from United Network for Organ Sharing (UNOS)
- Two ways to get a new kidney: 1) match with a deceased organ donor, or 2) find a living donor who matched with you (typically a family member or friend)

Our running example today

Kidney exchange algorithms

Roth and colleagues realized there was lost potential in the family member or friend who is willing to donate a kidney, yet isn't a match for the person in need

A cyclic exchange involving two patient-donor pairs

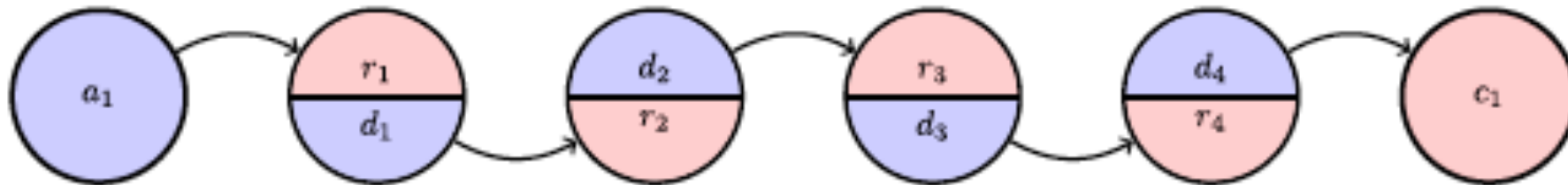


Anderson, R., Ashlagi, I., Gamarnik, D., & Roth, A. E. (2015). Finding long chains in kidney exchange using the traveling salesman problem. *Proceedings of the National Academy of Sciences*, 112(3), 663-668.

Our running example today

Kidney exchange algorithms

A chain exchange involving an altruistic donor, four patient–donor pairs, and a patient with no donor



Anderson, R., Ashlagi, I., Gamarnik, D., & Roth, A. E. (2015). Finding long chains in kidney exchange using the traveling salesman problem. *Proceedings of the National Academy of Sciences*, 112(3), 663-668.

Our running example today



Our running example today

Finding long chains using traveling salesman problem

Problem: Patients with incompatible donors

Solution: **Automated algorithmic solution** that creates long donor chains

Result: Thousands of additional kidney transplants

Kidney exchange is one of the **clearest real-world successes of algorithmic design**

2. Social problem

How can we do such responsible and ethical AI with positive social impacts?

Step 1: Identify a social problem

- Not every interesting question is a social problem

Why do people like certain movies more than others?

Many people aren't healthy

Is social media good or bad for society?

Will AI ever truly understand humans?

How can we do such responsible and ethical AI with positive social impacts?

Step 1: Identify a social problem

- Not every interesting question is a social problem

~~Many people aren't healthy~~

Chronic illness reduces the number of years people can live and work in good health.

~~Is social media good or bad for society?~~

Heavy social media use causes body image concerns among adolescents.

What makes a problem well-defined?

A good **social problem statement** is:

- Observable
- Measurable
- Actionable
- Consequential
- Doesn't pre-specify a solution

What makes a problem well-defined?

Kidney exchange example

Problem: **Many patients needing kidney transplants cannot receive one despite the existence of willing living donors.**

Measurable: # on the wait list

What makes a problem well-defined?

Kidney exchange example

Problem: **Many patients needing kidney transplants cannot receive one despite the existence of willing living donors.**

Measurable: # on the wait list

Alternative problems:

1. The kidney transplant system is inefficient. (too vague)
2. The problem is incompatible blood types. (too broad)
3. The problem is that we need better algorithms for kidney exchange.
4. The problem is that kidney exchange chains are computationally hard to find.
5. Hospitals need a traveling salesman algorithm to find long kidney chains.
(computational problems, not social problems, i.e., what's wrong with solutions, not what the problem is)

Think–Pair–Share (5 min)

- Define a **social problem you personally care about** that AI might address
- Write:

Problem

Why it matters

Why it is measurable

This problem will be your personal running example throughout the lecture (you can refine)

Make your own copy and draft your notes:

<https://docs.google.com/document/d/1wNI5TbbRueK72Rcl-ClEr43m4k-k85tMR51aRceLfrg/edit?usp=sharing>

3. Stakeholders

Stakeholders

We need to ask ourselves:

- Who is affected?
- Who is doing the practice?
- Who has power?

Stakeholders

- Direct users
- Institutions
- Regulators
- Affected communities

Kidney exchange running example

- Patients
- Donors
- Hospitals
- Insurance companies
- National transplant registries
- Regulators

Why stakeholders matter?

Stakeholders shape:

- incentives
- constraints
- adoption
- fairness outcomes

Think-Pair-Share (5 min)

For your problem:

- List **at least five stakeholders**
- Write your stakeholder map for different roles
- Make them as specific as possible

4. ToC

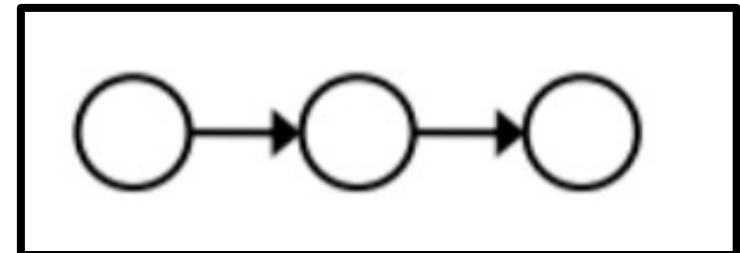
Theory of Change

- A theory of change specifies
 - How an **intervention** produces **outcomes**.

Theory of change

If X happens,
Then Y changes,
Which leads to Z impact

What is the causal chain?



Theory of change

Kidney exchange problem:

If algorithmic matching identifies chains

→ more compatible matches

→ more kidney transplants

Theory of change

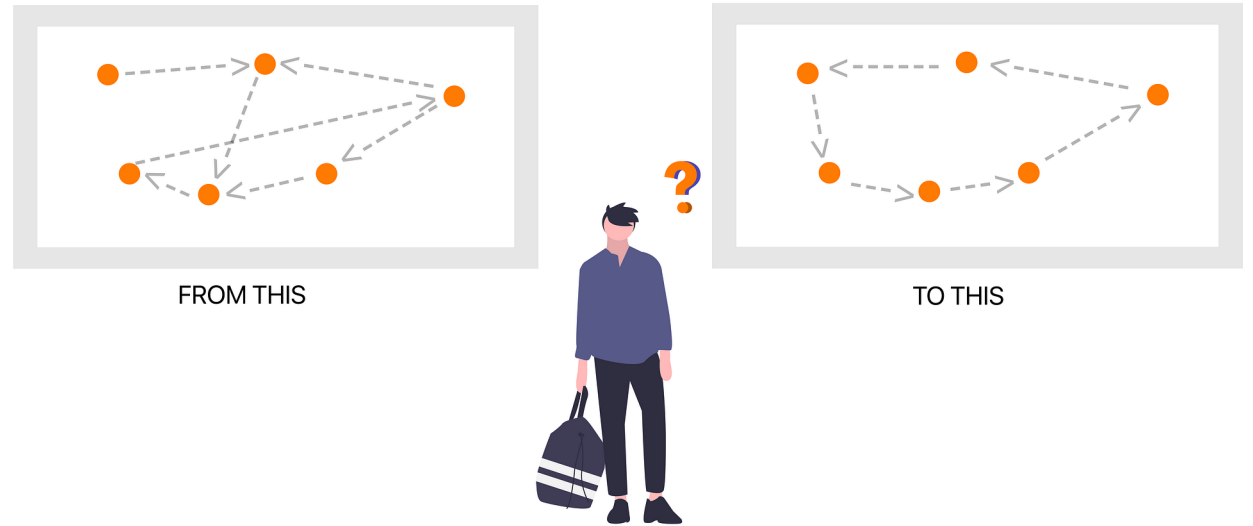
Kidney exchange problem:

The algorithm matters here

Manual matching vs. optimization algorithms

Traveling salesman formulation:

finding the shortest possible route that visits cities exactly once, and returns to the origin city



Theory of change

Kidney exchange problem:

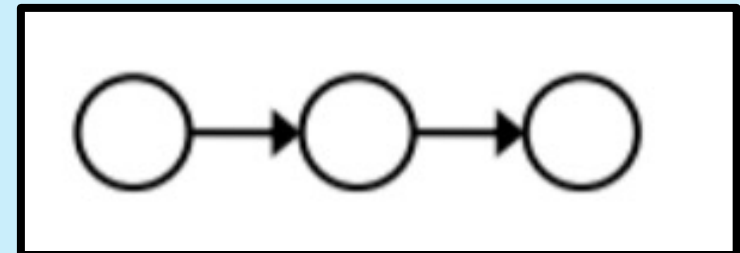
- Chains are computationally complex, so optimization enables **longer chains!**
- **And longer chains** -> more lives saved

If causal link unclear, AI applications fail

Think-Pair-Share (5 min)

- For your problem, in your document,
- Write a theory of change:
- If we introduce AI tool **X**
then **Y** changes
leading to outcome **Z**

What is the causal chain?



5. Institutions and partners

Institutions and partners

- Technology alone rarely solves social problems.
 - Institutions matter; they provide
 - infrastructure
 - authority
 - coordination
- > Scaling up the impact

Institutions and partners

Kidney exchange

Hospitals

UNOS

National kidney registries

- Hospitals originally reluctant to share donors
- Design system compatible with incentives

Institutions and partners

Scaling!

From small exchanges

→ to **national kidney exchange programs!**

Institutions and partners

How do we find them?

- Google, LinkedIn, Google Scholar, etc.
- Word of mouth & weak ties

Think-Pair-Share (5 min)

For your AI intervention: Which institutions must be involved to scale this up and have positive impact? List at least 5 specific ones:

- Universities
- Companies
- Public sector
- Government
- Non-profits
- ?? Something else?

Make it as specific as possible! Search for specific players who could be your partners, specific organizations, specific people (Google, LinkedIn, etc.)

6. Evaluation

Evaluation

- How do we know if the intervention works?

Outcome metrics

Behavioral change

Equity impact

- Why metrics matter?

Without metrics: impact cannot be assessed!

Evaluation

Kidney exchange

- Chain length
- Number of transplants
- Waiting time

7. Risks

Risks

- Every intervention introduces risks
- Risks we talked about so far can emerge:
 1. Unethical development and experimentation (justice, consent, respect)
 2. Surveillance
 3. Data and annotation issues (harms, compensation, invisibility)
 4. User safety risks
 5. Fairness metrics violated across groups
 6. Stereotypes amplified

Risks

Kidney exchange risks

- Chain collapse
- Fairness concerns
- Logistical failure

Think-Pair-Share

- Define **one key success metric** for your idea.
 - Metric and a specific value or percentage change you would be targeting
- Define **three risks** you're most worried about
 - How would they be identified?
 - How could they be mitigated?

Class concepts

Unethical development and experimentation
(justice, consent, respect) (Lec 2)
Surveillance (Lec 2)
Data and annotation issues (harms, compensation, invisibility) (Lec 2)
User **safety risks** (Lec 3)
Fairness metrics violated across groups (Lec 5)
Stereotypes amplified (Lec 6)

Post-class exercise (optional)

- Clean up and convert your notes document to a 1-page executive summary document with structured headers
- Upload the PDF file of your personal Theory of Change / a mini policy design exercise template
- This is optional, but recommend so that you have it for your own record; this can be a framework of applying AI ethically that you may reuse in other contexts

Framework for Ethical AI Deployment

1. Define the social problem
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Wednesday

- We have three readings where authors have different visions for how AI should be developed
- The first reading presents a number of tangible problems & AI solutions that we'll discuss

Shaping AI's Impact on Billions of Lives

[Mariano-Florentino Cuéllar](#), [Jeff Dean](#), [Finale Doshi-Velez](#), [John Hennessy](#), [Arman](#), [Sanmi Koyejo](#), [Pelonomi Moiloa](#), [Emma Pierson](#), and [David Patterson](#)

Article | [Open access](#) | Published: 01 September 2024

AI Governance in a Complex and Rapidly Changing Regulatory Landscape: A Global Perspective

[Esmat Zaidan](#) ✉ & [Imad Antoine Ibrahim](#)

[Humanities and Social Sciences Communications](#) 11, Article

Participation is not a Design Fix for Machine Learning

Mona Sloane ^{*1} Emanuel Moss ^{*2} Olaitan Awomolo ^{*3} Laura Forlano ^{*4}

Feedback form—please use it!

<https://tinyurl.com/4sfsjszu>



References

- [Anderson, R., Ashlagi, I., Gamarnik, D., & Roth, A. E. \(2015\). Finding long chains in kidney exchange using the traveling salesman problem. Proceedings of the National Academy of Sciences, 112\(3\), 663-668.](#)

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